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 Studierichting: Informatica
 Oefeningenreeks 6C nr. 4.6

$$x_n = \sqrt{n^2 + 3n} - n$$

$$\begin{aligned} \lim_{n \rightarrow +\infty} x_n &= \lim_{n \rightarrow +\infty} \sqrt{n^2 + 3n} - n \\ &= \lim_{n \rightarrow +\infty} (\sqrt{n^2 + 3n} - n) \cdot \frac{\sqrt{n^2 + 3n} + n}{\sqrt{n^2 + 3n} + n} \\ &= \lim_{n \rightarrow +\infty} \frac{n^2 + 3n - n^2 + n\sqrt{n^2 + 3n} - n\sqrt{n^2 + 3n} - n^2}{\sqrt{n^2 + 3n} + n} \\ &= \lim_{n \rightarrow +\infty} \frac{3n}{\sqrt{n^2 + 3n} + n} \\ &= \lim_{n \rightarrow +\infty} \frac{3n}{n\sqrt{1 + \frac{3}{n}} + n} \\ &= \frac{3}{1 + 1} = \frac{3}{2} \\ \text{adh}_{n \rightarrow +\infty} x_n &= \left\{ \frac{3}{2} \right\} \\ \liminf_{n \rightarrow +\infty} x_n &= \frac{3}{2} \\ \limsup_{n \rightarrow +\infty} x_n &= \frac{3}{2} \end{aligned}$$

References